61. Which of the following imparts green colour to the glass

(a) Cu_2O
(b) CdS
(c) MnO_2
(d) Cr_2O_3

Answer: D — (d) Cr_2O_3

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in Cr_2O_3 has the characteristic oxidation state/ligand field that produces the observed colour.

62. On the heating copper nitrate strongly, ... is finally obtained

(a) Copper
(b) Copper oxide
(c) Copper nitrate
(d) Copper nitride

Answer: B — (b) Copper oxide

Chemical reason: On strong heating, copper(II) nitrate decomposes to copper(II) oxide with NO_2 and O_2: $2Cu(NO_3)_2 \rightarrow 2CuO + 4NO_2 + O_2$. Hence CuO is the final solid.

63. On adding KI to a solution of copper sulphate

(a) Cupric oxide is precipitated
(b) Metallic copper is precipitated
(c) Cuprous iodide is precipitated with liberation of iodine
(d) No change occurs

Answer: C — (c) Cuprous iodide is precipitated with liberation of iodine

Chemical reason: Cu^{2+} oxidizes I^- to I_2 and is reduced to Cu^+, which precipitates as CuI: $2Cu^{2+} + 4I^- \rightarrow 2CuI(s) + I_2$. Therefore CuI and iodine are formed.

64. Which of the following statements is correct about equivalent weight of $KMnO_4$

(a) It is one third of its molecular weight in alkaline medium
(b) It is one fifth of its molecular weight in alkaline medium
(c) It is equal to its molecular weight in acidic medium
(d) It is one third of its molecular weight in acidic medium

Answer: A — (a) It is one third of its molecular weight in alkaline medium

Chemical reason: Equivalent weight is calculated as molar mass divided by the number of electrons exchanged per formula unit in the stated redox reaction.

65. The reaction of $K_2Cr_2O_7$ with NaCl and conc. H_2SO_4 gives

(a) $CrCl_3$

(b) $CrOCl_2$

(c) CrO_2Cl_2

(d) Cr_2O_3

Answer: C — (c) CrO_2Cl_2

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives CrO_2Cl_2. The choice is consistent with the expected oxidation state and stability of the product.

66. Silver nitrate is supplied in coloured bottles because it is

(a) Oxidised in air

(b) Decomposes in sunlight

(c) Explosive in sunlight

(d) Reactive towards air in sunlight

Answer: B — (b) Decomposes in sunlight

Chemical reason: $AgNO_3$ is photosensitive. Light causes slow photochemical decomposition/reduction, so it is stored in amber or coloured bottles that limit light exposure.

67. A nitrate when mixed with common salt gives a white precipitate which is soluble in dilute NH_4OH. It is the nitrate of

(a) Copper

(b) Mercury

(c) Silver

(d) Gold

Answer: C — (c) Silver

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives Silver. The choice is consistent with the expected oxidation state and stability of the product.

68. Which one of the following is lunar caustic

(a) $AgNO_3$

(b) Cu_2Cl_2

(c) $CuCl_2$

(d) Hg_2Cl_2

Answer: A — (a) $AgNO_3$

Chemical reason: Fused silver nitrate is historically called lunar caustic because of its caustic action and association of silver with the moon (“luna”).

69. Invar, an alloy of Fe and Ni is used in watches and meter scale, its characteristic property is

(a) Small coefficient of expansion

(b) Resistance to corrosion

(c) Hardness and elasticity

(d) Magnetic nature

Answer: A — (a) Small coefficient of expansion

Chemical reason: Invar is a Fe–Ni alloy with an exceptionally small coefficient of thermal expansion, so its dimensions change very little with temperature; this is why it is useful in precision instruments.

70. The extraction of nickel involves

(a) The formation of $Ni(CO)_4$

(b) The decomposition of $Ni(CO)_4$

(c) The formation and thermal decomposition of $Ni(CO)_4$

(d) The formation and catalytic decomposition of $Ni(CO)_4$

Answer: C — (c) The formation and thermal decomposition of $Ni(CO)_4$

Chemical reason: The correct choice is The formation and thermal decomposition of $Ni(CO)_4$. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

71. On adding excess of NH_3 solution to $CuSO_4$ solution, the dark blue colour is due to

(a) $Cu(NH_3)_4+^+$

(b) $Cu(NH_3)_2+^+$

(c) $Cu(NH_3)^+$

(d) None of the above

Answer: A — (a) $Cu(NH_3)_4$^+$^+$

Chemical reason: Excess NH_3 added to Cu^{2+} forms the intensely deep-blue tetraammine copper(II) complex, commonly written $[Cu(NH_3)_4(H_2O)_2]^{2+}$. The colour therefore comes from complex formation around Cu^{2+}.

72. If M is the molecular weight of $KMnO_4$, its equivalent weight will be when it is converted into K_2MnO_4

(a) M

(b) $M/_3$

(c) $M/_5$

(d) $M/_7$

Answer: A — (a) M

Chemical reason: Equivalent weight is calculated as molar mass divided by the number of electrons exchanged per formula unit in the stated redox reaction.

73. While writing the formula of ferrous oxide it is written as (FeO), because it is

(a) Non-stoichiometric

(b) Non-existant

(c) Paramagnetic

(d) Ferromagnetic

Answer: A — (a) Non-stoichiometric

Chemical reason: Wüstite is non-stoichiometric iron(II) oxide, commonly written $Fe_{1-x}O$ rather than perfectly FeO, because some Fe^{2+} sites are vacant and Fe^{3+} is present for charge balance.

74. Which of the following exhibit maximum oxidation state of vanadium

(a) $VOCl_3$

(b) VCl_4

(c) VCl_3

(d) VCl_2

Answer: A — (a) $VOCl_3$

Chemical reason: The result is obtained by assigning oxidation numbers before and after the reaction and balancing electron transfer. The species in $VOCl_3$ has the oxidation-state change required by the stated redox conditions, so that choice is correct.

75. Prussian blue is due to the formation of

(a) $Fe_4Fe(CN)_{63}$

(b) $Fe_2Fe(CN)_6$

(c) $Fe_3Fe(CN)_6$

(d) $FeFe(CN)_{63}$

Answer: A — (a) $Fe_4Fe(CN)_{63}$

Chemical reason: Prussian blue is produced by ferric ions with ferrocyanide, forming a mixed-valence iron cyanide network often represented as $Fe_4[Fe(CN)_6]_3{\cdot}xH_2O$.

76. The Nesseler's reagent contains

(a) Hg_2+^+

(b) $Hg+^+$

(c) HgI_2-^-

(d) HgI_4-^-

Answer: D — (d) HgI_4-^-

Chemical reason: Nessler's reagent is an alkaline solution of potassium tetraiodomercurate(II), $K_2[HgI_4]$, used for detecting ammonia. The complex mercury iodide species is the essential reagent component.

77. Formula of ferric sulphate is

(a) $FeSO_4$

(b) $Fe(SO_4)_2$

(c) Fe_2SO_4

(d) $Fe_2(SO_4)_3$

Answer: D — (d) $Fe_2(SO_4)_3$

Chemical reason: Ferric sulfate contains Fe^{3+} and SO_4^{2-}. Charge balance requires 2 Fe^{3+} (+6) with 3 sulfate ions (−6), giving $Fe_2(SO_4)_3$.

78. When $CuSO_4$ is hydrated, then it becomes

(a) Acidic

(b) basic

(c) Neutral

(d) Amphoteric

Answer: D — (d) Amphoteric

Chemical reason: The correct choice is Amphoteric. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

79. Silvering of mirror is done by

(a) $AgNO_3$

(b) Ag_2O_3

(c) Fe_2O_3

(d) Al_2O_3

Answer: A — (a) $AgNO_3$

Chemical reason: Silver chemistry is governed strongly by low-solubility halides and complex formation with ligands such as NH_3 or $S_2O_3^{2-}$.

Under the stated conditions this leads to $AgNO_3$.

80. The colour of $K_2Cr_2O_7$ changes from red orange to lemon yellow on treatment with aqueous KOH because of

(a) The reduction of CrVI to CrIII

(b) The formation of chromium hydroxide

(c) The conversion of dichromate to chromate

(d) The oxidation of potassium hydroxide to potassium peroxide

Answer: C — (c) The conversion of dichromate to chromate

Chemical reason: Adding OH^- shifts the chromate–dichromate equilibrium toward chromate: $Cr_2O_7^{2-} + 2OH^- \rightleftharpoons 2CrO_4^{2-} + H_2O$. Dichromate is orange‑red whereas chromate is yellow.